

Predictive Analytics PS- 7

Shreyanshi Adak - 745

2026-03-19

#Problem Set 7: Shrinkage Methods

Load Data and Preprocess

```
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.5.2

library(glmnet)

## Warning: package 'glmnet' was built under R version 4.5.2
## Loading required package: Matrix
## Warning: package 'Matrix' was built under R version 4.5.2
## Loaded glmnet 4.1-10

# Load and clean data
data(Hitters)
Hitters <- na.omit(Hitters)

# Create model matrix (dummy variables)
x <- model.matrix(Salary ~ ., Hitters)[, -1]
y <- Hitters$Salary
```

Q1. Split the data set into training (remaining) and test set (100 observations)

```
set.seed(123)
train_index <- sample(1:nrow(x), nrow(x) - 100)

x_train <- x[train_index, ]
y_train <- y[train_index]

x_test <- x[-train_index, ]
y_test <- y[-train_index]
```

Q2. Train least squares model and report train MSE and test MSE

```
# Fit least squares
lm_fit <- lm(Salary ~ ., data = Hitters, subset = train_index)

# Predictions
train_pred <- predict(lm_fit, newdata = Hitters[train_index, ])
test_pred <- predict(lm_fit, newdata = Hitters[-train_index, ])
```

```
# MSE
train_mse <- mean((y_train - train_pred)^2)
test_mse <- mean((y_test - test_pred)^2)
```

```
train_mse
## [1] 73584.77

test_mse
## [1] 146030.2
```

Q3. Ridge Regression

```
# Grid of Lambda
lambda_grid <- 10^seq(10, -2, length = 100)

ridge_fit <- glmnet(x_train, y_train, alpha = 0, lambda = lambda_grid)

dim(coef(ridge_fit))
## [1] 20 100
```

(ii) L2 norm for 50th and 60th lambda

```
coef_50 <- coef(ridge_fit)[,50]
coef_60 <- coef(ridge_fit)[,60]

l2_50 <- sqrt(sum(coef_50[-1]^2)) # exclude intercept
l2_60 <- sqrt(sum(coef_60[-1]^2))

l2_50
## [1] 9.083952

l2_60
## [1] 70.87237
```

Comment: L2 norm decreases as λ increases $\rightarrow$ coefficients shrink.

(iv) Test MSE for λ_{50} and λ_{60}

```
pred_50 <- predict(ridge_fit, s = lambda_grid[50], newx = x_test)
pred_60 <- predict(ridge_fit, s = lambda_grid[60], newx = x_test)

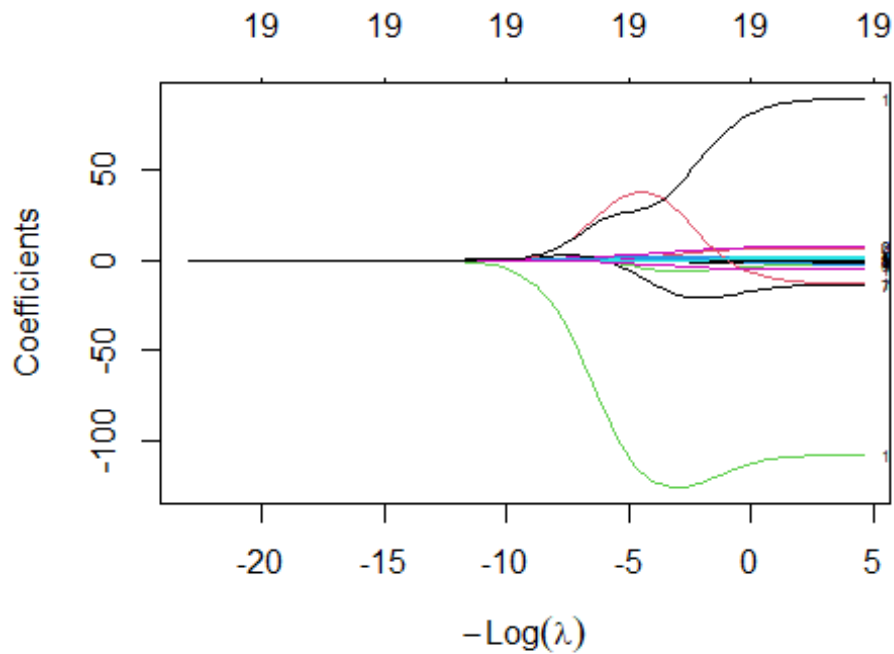
mse_50 <- mean((y_test - pred_50)^2)
mse_60 <- mean((y_test - pred_60)^2)

mse_50
## [1] 135323.5
```

```
mse_60  
## [1] 126500.1
```

Q4. Plot standardized coefficients vs λ

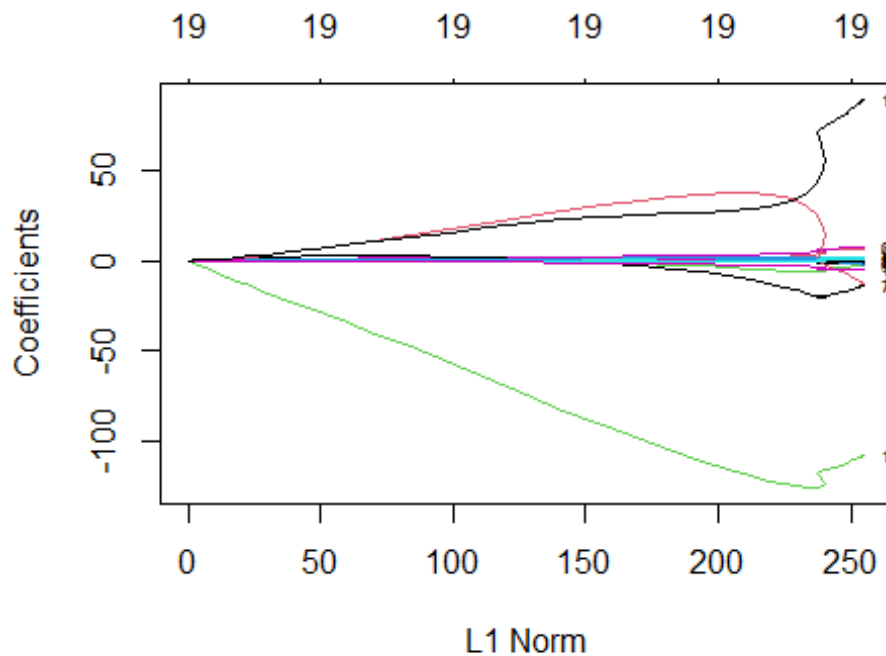
```
plot(ridge_fit, xvar = "lambda", label = TRUE)
```



Interpretation: As λ increases, coefficients shrink toward zero but never become exactly zero.

Q5. Plot vs ratio of L2 norms

```
plot(ridge_fit, xvar = "norm", label = TRUE)
```



Interpretation: Graph becomes smoother; x-axis now shows shrinkage relative to least squares instead of λ .

Q6. Ridge with Cross-Validation

```
set.seed(123)
cv_ridge <- cv.glmnet(x_train, y_train, alpha = 0)

best_lambda <- cv_ridge$lambda.min

# Test MSE
ridge_pred <- predict(cv_ridge, s = best_lambda, newx = x_test)
ridge_mse <- mean((y_test - ridge_pred)^2)

best_lambda
## [1] 31.20346

ridge_mse
## [1] 139188.6
```

Q7. LASSO with Cross-Validation

```
set.seed(123)
cv_lasso <- cv.glmnet(x_train, y_train, alpha = 1)
```

```

lasso_pred <- predict(cv_lasso, s = "lambda.min", newx = x_test)
lasso_mse <- mean((y_test - lasso_pred)^2)

# Number of non-zero coefficients
lasso_coef <- coef(cv_lasso, s = "lambda.min")
nonzero <- sum(lasso_coef != 0) - 1 # exclude intercept

lasso_mse

## [1] 142142.6

nonzero

## [1] 15

```

Q8. Elastic Net with Cross-Validation

```

set.seed(123)
cv_enet <- cv.glmnet(x_train, y_train, alpha = 0.5)

enet_pred <- predict(cv_enet, s = "lambda.min", newx = x_test)
enet_mse <- mean((y_test - enet_pred)^2)

# Non-zero predictors
enet_coef <- coef(cv_enet, s = "lambda.min")
enet_nonzero <- rownames(enet_coef)[enet_coef[,1] != 0]

enet_mse

## [1] 141649.7

enet_nonzero

## [1] "(Intercept)" "AtBat" "Hits" "HmRun" "RBI"
## [6] "Walks" "Years" "CAtBat" "CHits" "CHmRun"
## [11] "CRuns" "CRBI" "CWalks" "LeagueN" "DivisionW"
## [16] "PutOuts" "Assists" "Errors" "NewLeagueN"

```

Q9. Comparison of Models

```

train_mse

## [1] 73584.77

test_mse

## [1] 146030.2

ridge_mse

## [1] 139188.6

lasso_mse

```

```
## [1] 142142.6
```

```
enet_mse
```

```
## [1] 141649.7
```

Comment:

Least Squares: High variance → higher test error

Ridge: Reduces variance → improves prediction

LASSO: Performs variable selection → sparse model

Elastic Net: Balance of Ridge & LASSO

Conclusion: Shrinkage methods (especially LASSO/Elastic Net) generally give better prediction accuracy for Salary.